Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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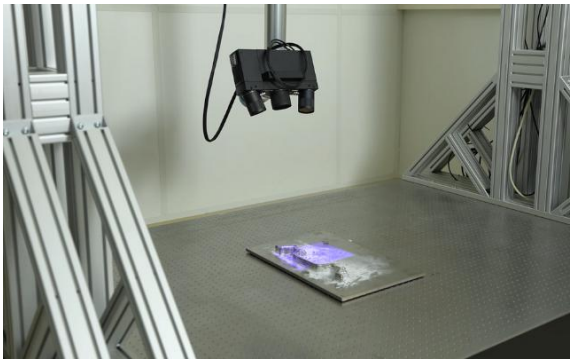
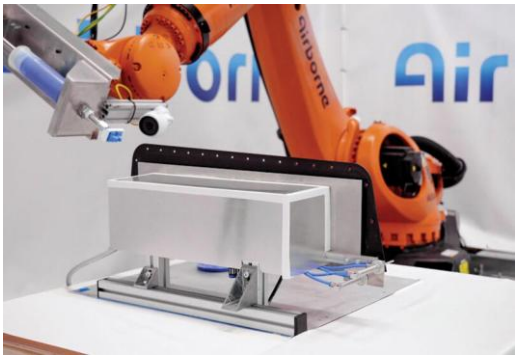
2. Technology Overview

Title of Proposed Technology	**Advanced Robotics for 3D Nesting: Precision Quality Control and Manufacturing Efficiency**  			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical/materials		*Electrical/Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Assembly/Precision Transfer Technologies, Measurement/Sensor Technologies for Automation*	*Measuring Instruments*	*Measurement Sensors and Components*

Concept of Proposed Technology	TNO's 3D Nesting technology automates the entire workflow from CAD/CAM to positioning and post-processing of complex, freeform components. Using advanced 3D cameras, point cloud analysis, and AI-driven alignment, the system enables reworking, edge sealing, and material restoration in a single, first-time-right step. Recent scientific studies show that vision-based robot manipulation, leveraging sensor feedback and machine learning, significantly improves robotic localization, orientation, and control—crucial for accurate 3D nesting. Combining 2D and 3D vision (e.g., TOF + RGB) enhances segmentation precision and reduces computation time by orders of magnitude, achieving sub-millimeter accuracy in realistic environments.
Rationale for Joint R&D Project	High-tech manufacturing (e.g., additive manufacturing and refurbishment) requires first-time-right precision, which is challenging for manual intervention in high-mix/low-volume scenarios. TNO's 3D Nesting reduces process steps by 10–30% and ensures consistent output quality. However, large-scale implementation in Korean SMEs requires joint R&D: • TNO provides cutting-edge AI, point-cloud alignment, NDT imaging, obotics and topology expertise. • Korean SMEs contribute manufacturing systems, robot platforms, and market access.
Key Words	1. Advanced Robotics 2. 3D Nesting 3. Quality Control 4. Manufacturing Efficiency 5. Intelligent Imaging 6. Automated workflow CAM/CAD – AM - CNC

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	*TRL 4 (Lab validation of integrated modules)*	**End of the Project**	*TRL 6 (Pilot demonstration in SME environment: Production & performance evaluation of prototypes at a pilot scale)*

R&D Objectives	***Main R&D objectives:*** 1. *Develop AI algorithms for automated CAD-to-reality alignment* 2. *Integrate robotics for autonomous positioning, edge sealing, and material restoration* 3. *Implement quality control using intelligent imaging and connection with ultrasound NDT (DVM)* 4. *Sensor Fusion* 5. *Define URS, SRS and prototype for testing and validation of the (business) cases.* The primary objective of the R&D trajectory is to mature TNO's 3D Nesting technology from **TRL 4 to TRL 6**, extending its capabilities in **AI-based alignment, sensor fusion, high-precision robotics, and automated quality control**. The project aims to establish an integrated system that automates the complete workflow from **CAD/CAM preparation** to **component positioning, post-processing, repair, and quality verification**. A key part of this effort is to deeply understand and formalize the needs of Korean SMEs working in **additive manufacturing, repair and refurbishment, precision machining, and high-mix/low-volume production**. These real-world use cases will form the foundation of the project's User Requirement Specification (URS) and System Requirement Specification (SRS), ensuring the resulting system is technically feasible, industrially deployable, and economically viable within Korean SME production environments. A central ambition of the R&D work is to transform TNO's 3D Nesting Lab solution into a **flexible, first-time-right manufacturing prototype module** that can be seamlessly integrated into both new and existing SME robotic production cells. This includes the development of advanced AI algorithms for **CAD-to-reality alignment**, **automated part registration**, **adaptive toolpath generation**, and **closed-loop correction** during machining or repair. Complementing this effort, the project will advance multimodal quality assurance based on **intelligent imaging, ultrasound NDT, and point-cloud analytics**, enabling automated defect detection and inline verification steps. Through the combination of autonomous manipulation, vision-based decision making, and integrated inspection, the system intends to drastically reduce manual intervention, process variability, cycle time, and rework. Building on TNO's extensive laboratory experience with 3D Nesting, the project will mature the current prototype system into a **robust, autonomous, and production-ready industrial solution**. Past lab experiments—including structured-light scanning of turbine blades, point-cloud alignment on fuselage panels, and corrosion-repair inspection workflows—demonstrated that robust autonomy in complex geometries requires: • advanced multi-sensor co-registration (images, point clouds, ultrasound) • robust alignment algorithms capable of handling incomplete or noisy data • digital-twin-based planning of robotic trajectories • integrated defect detection and thickness-analysis capabilities • a user-centric data workbench for visualization, decision support, and traceability The objective of the proposed R&D effort is therefore to consolidate these lab-validated capabilities into a **scalable, modular, and industrial architecture**. The system must be capable of automatically adapting AI models, scanning routines and inspection logic to a wide range of component types, without requiring manual reprogramming, so Korean SMEs can efficiently deploy the solution for AM post-processing, CNC finishing, hybrid manufacturing, repair workflows, and precision alignment tasks. Throughout the project, all activities will be structured using a **systems-engineering methodology**, ensuring rigorous traceability from user needs to system functionality, architecture, and performance validation. This guarantees that the technology meets both operational and economic requirements and that scalability and repeatability are embedded from the outset. In addition, the

	R&D will investigate how **topology optimization** and **Industry 4.0 methodologies** can support the business case particularly for SMEs transitioning towards lightweighting, structural redesign, advanced repair strategies, or digital-factory integration where 3D Nesting is a key enabler. To achieve these objectives, the R&D trajectory is organized into four work packages: **WP1 – User Requirements Specification (URS)** Use-case elicitation from participating Korean SMEs, including formalization of operational, technical, accuracy, sustainability, safety, and data-handling needs. **WP2 – System Requirement Specification (SRS)** Translation of URS findings—augmented with performance benchmarks from TNO's lab research—into detailed specifications for scanning hardware, AI modules, system architecture, robot integration, inspection subsystems, and automation logic. **WP3 – Prototype Development** Integration of TNO's proven modules (structured light scanning, point-cloud processing, alignment engine, inspection modules) into a unified 3D Nesting demonstrator suitable for real industrial components. **WP4 – Test & Validation, CAPEX/OPEX Assessment** Pilot-scale deployment in Korean SME environments to validate cycle time, throughput, accuracy, robustness, maintenance needs, and to derive a quantified business case including CAPEX/OPEX modelling and ROI analysis. If beneficial within the SME business case, **topology-optimization support** will be integrated as an optional module, allowing component redesign, repair strategy optimization, and structural improvement directly informed by 3D Nesting's measurement and alignment data.
R&D Target Specifications	The R&D target specifications define the key performance parameters required to ensure successful industrial deployment of the 3D Nesting system. These specifications will be further detailed and refined based on the specific production scenarios, geometries, materials, and throughput requirements of participating Korean SMEs. This specification framework ensures a clear alignment between user expectations (URS), system functions (SRS), and measurable performance criteria during prototype development, validation, and pilot deployment. At the **core of the system performance objectives** are **four key targets**: **(1)** alignment accuracy <0.1 mm **(2)** cycle time <20 s per component, **(3)** defect detection ≥95% (visual + ultrasound + 3D point-cloud modalities), and **(4)** adaptability for ≥100 component types without manual reprogramming. From a technical performance perspective, the system aims to achieve CAD-to-reality alignment accuracy below 0.1 mm, enabling high-precision placement, repair, edge sealing, hybrid AM-CNC finishing, and other operations on complex freeform geometries. Cycle-time targets are set to <20 seconds per component for alignment and positioning, ensuring compatibility with industrial takt times in high-mix/low-volume manufacturing. Inspection capabilities must reach ≥95% defect detection accuracy across combined modalities—including visual imaging, structured-light scanning, point-cloud analysis, and non-contact ultrasound NDT—to detect cracks, porosity, waviness, corrosion thinning, delamination, and other critical defect types. To support flexible deployment across diverse manufacturing environments, the system will be designed to

autonomously handle at least 100 unique component types without manual reprogramming, leveraging AI-driven generalization, adaptive scanning routines, and parametric process templates.

1. Alignment & Accuracy Requirements

Laboratory experiments (TRL 4) have shown that sub-millimeter precision is achievable through integrated co-registration of structured-light 3D scanning, point-cloud data and ultrasound-based measurements. The TRL 6 objective is to industrialize and robustify these capabilities: CAD-to-reality alignment accuracy <0.1 mm (goal <0.05 mm for small, rigid components); stable alignment performance on shiny, reflective, curved, complex, and freeform geometries used in mobility, aerospace, semiconductor tooling, and precision machinery. These accuracy requirements enable reliable first-time-right positioning, automated finishing, and high-quality repair workflows.

2. Cycle Time & Throughput Requirements

TRL 4 laboratory setups demonstrate high-speed scanning and computation; the TRL 6 system must meet industrial takt-time expectations: automated alignment + nesting cycle <20 seconds per component (goal <10 seconds for small geometries); end-to-end automated inspection time reduced by a factor 3–5 compared to manual workflows. These targets ensure competitiveness in high-mix/low-volume SME environments where throughput and manpower efficiency are critical.

3. Sensor Fusion & Inspection Requirements

To support AM post-processing, refurbishment, precision machining and structural repair, the TRL 6 system will integrate multimodal sensing: defect detection ≥95% across visual imaging, structured-light 3D scanning, point-cloud analytics and ultrasound thickness mapping; automated thickness and geometry mapping using non-contact ultrasound or equivalent NDT methods; integrated co-registration engine capable of overlaying scans, point clouds, photos and technical drawings into one unified coordinate system for automated decision-making and reporting.

4. Adaptability & Automation Requirements

To enable SMEs to operate the system across diverse product portfolios: autonomous handling of ≥100 component types without manual reprogramming, supported by AI-based shape recognition and adaptive routines; automatic generation of golden reference models from multiple scans when CAD is unavailable; digital-twin-enabled robot path planning for positioning, machining, repair or hybrid manufacturing workflows. These adaptability targets ensure scalability across multiple industrial sectors.

5. System Integration & Deployment Requirements

The TRL 6 prototype will be engineered for seamless integration into SME production flows: modular interfacing with AM, CNC, repair and hybrid manufacturing equipment; automated reporting, traceability and data management for audits and regulatory compliance; pilot-scale validation in SME environments (TRL 6), including assessments of robustness, ergonomics, maintainability, uptime and environmental performance.

6. Economic & Operational Performance Targets (CAPEX/OPEX)

The TRL 4 research platform already indicates meaningful efficiency gains. The TRL 6 deployment will target: 20–40% reduction in manual processing steps through automated alignment, inspection and decision-support; <5% rework rate enabled by improved accuracy, repeatability and inline defect detection; clear ROI improvements via increased throughput, scrap reduction, shorter changeover times and lower operator workload. These economic targets will be validated during WP4 pilot deployments and used to establish a robust business case for Korean SME adoption. If required, the R&D team can support additional activities such as topology optimization, simulation-based structural design, or automated repair-strategy generation to strengthen the value

	proposition and support broader commercialization.
R&D Plan	**High level Preliminary planning to be up dated based on the specific use cases:** • **Month 0-6:** Define User Requirement Specification and System requirement specification based on SME use-cases. • **Month 6-18:** develop and extend existing lab prototype with System architecture, AI model development, 3D vision module integration, 3D nesting integration, robotic handling integration incl. preliminary test runs • **Month 18-28:** Pilot demonstration, performance testing and validation, • **Month > 28:** technology & commercialization roadmap development and industrial implementation at participating SME's

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Develop AI vision & point-cloud alignment; • integrate camera and nesting technologies; • Prototype testbed; • system integration and validation; • training & knowledge transfer.
Expected R&R of Korean SME Partner	• Provide manufacturing use cases; • Integrate prototype setup within AM-CNC or other production cell • Test & fine-tune in pilot prototype; • Market introduction in Korea.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• **Technological**: Unique integration of 3D nesting, robotics, with continuous Machine Learning optimization • **Commercial:** Accelerated adoption of automated post-processing and refurbishment in high-tech SMEs • **Economic:** Higher productivity, lower costs, improved quality, and increased competitiveness for Korean and European SMEs + option of licensing core technology.

6. References

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5. **AAS & RDF Modelling in Industry 4.0 – TNO**
   https://publications.tno.nl/publication/34641154/LyQGGI/rongen-2023-modelling.pdf
   [publications.tno.nl]
6. **Patents / IP**

   - PLT2020075 Non-contact inspection technique using MEMs-microphones
   - PLT2023031 Non-contact inspection technique for anisotropic composites
   - PLT2023046 Patent application 3D nesting
   - I-Depot visual robot feedback for tracking and picking
   - PLT2017057 User assistance in assembly – operator work station